Construction Chemicals

Product Datasheet
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Waterproofing

O-Swell

Hydrophilic Swelling Waterstop
for Construction Joints



Product Description

O-Swell is a German-technology hydrophilic swelling waterstop strip based on synthetic rubber (butyl / bentonite). On contact with water it expands by 1000 – 1300 % and seals construction joints, pipe penetrations and gaps in concrete. It re-swells after drying, needs no welding and is suitable for fresh, saline and waste water.

Uses

- Horizontal and vertical construction joints.
- Pipe penetrations through concrete walls and slabs.
- Joints between precast elements.
- Basements, tanks, tunnels and foundations.
- Structures exposed to sea water or sewage.

Advantages

- Swells 1000 – 1300 % on contact with water.
- Seals against water pressure up to 6 bar.
- No welding needed; quick installation.
- Re-swells after repeated wet/dry cycles.
- Suitable for saline and waste water.

Application

1. **Surface Preparation**
   Clean, sound and even concrete surface at the joint; remove laitance and loose particles.
2. **Fixing**
   Fix with concrete nails every ~300 mm, in a profile channel, or bond with adhesive; butt-join ends without gaps.
3. **Concreting**
   Keep minimum 60 mm concrete cover; pour the next stage before the strip is exposed to water.

Consumption

1 m per 1 m of joint
plus butt-joint overlap
For pipe penetrations, one full turn around the pipe.

Packaging

Rolls of 10 m; 5 rolls per carton. Sizes 20 × 5 mm and 20 × 10 mm.

Storage & Shelf Life

Store in original cartons in a dry place, protected from water, rain and humidity, at +5 °C to +35 °C.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Material	Hydrophilic synthetic rubber (butyl / bentonite)	—
Free swelling in water	1000 – 1300 %	—
Swelling in rain water	1100 %	—
Swelling in concrete pore water	950 %	—
Hardness	Shore A 35	—
Tensile strength	> 2 MPa	—
Elongation	> 400 %	—
Water pressure resistance	up to 6 bar	—
Service temperature	-20 °C to +75 °C	—
Minimum concrete cover	60 mm	—

Consumption

Size (width × thickness)	Roll length	Pack
20 × 5 mm	10 m	5 rolls / carton
20 × 10 mm	10 m	5 rolls / carton

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- For construction joints only; not for expansion or moving joints.
- Minimum concrete cover over the strip: 60 mm.
- Substrate clean, sound and even.
- Protect the strip from water and rain before concreting to prevent premature swelling.
- Pour the next concrete stage as soon as possible after installation.

Application Notes

- Fix with concrete nails approximately every 300 mm, in a profile channel, or with a compatible adhesive.
- Butt-join strip ends tightly; do not leave gaps.
- On pipe penetrations, wrap fully around the pipe and secure with tie wire.
- Install in the centre of the joint where possible.
- Do not install in standing water.

Health & Safety

- Wear gloves when handling and cutting.
- Use safety glasses when nailing.
- Follow site safety rules for work in excavations.

First Aid & Cleaning

- Eyes: rinse with clean water.
- Skin: wash with soap and water.
- Cuts: clean and dress the wound.
- Clean tools with a dry cloth; no cleaning agent required.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.